

5.1 Two coupled Oscillators

Consider a system consisting of two equal masses, $m_1 = m_2 = m$, moving in a horizontal plane. These masses are coupled by a spring with spring constant k_{12} . Each mass is also attached to a spring with spring constant k , where the other ends of the springs are fixed to rigid walls. Let x_1 and x_2 represent the displacements of the masses from their equilibrium positions. With reference to the figure [10](#), the Lagrangian of the system is given by:

$$L = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 - \frac{1}{2}kx_1^2 - \frac{1}{2}k_{12}(x_2 - x_1)^2 - \frac{1}{2}kx_2^2. \quad (66)$$

The equation of motions of the two masses are

$$m\ddot{x}_1 + (k + k_{12})x_1 - k_{12}x_2 = 0, \quad m\ddot{x}_2 + (k + k_{12})x_2 - k_{12}x_1 = 0. \quad (67)$$

Before we try to solve the equations, notice that they can be written in a compact form as

$$\mathbf{M}\ddot{\mathbf{X}} + \mathbf{K}\mathbf{X} = 0, \quad (68)$$

where square matrices $\mathbf{M}$ and $\mathbf{K}$ are

$$\mathbf{M} = \begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} k + k_{12} & -k_{12} \\ -k_{12} & k + k_{12} \end{pmatrix},$$

and the column vector $\mathbf{X}$ is

$$\mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

Mathematically, the equation [\(68\)](#) is a generalization of the SHO equation of motion $m\ddot{x} + kx = 0$ such that the system have a dof $\mathbf{X}$ with mass $\mathbf{M}$ and spring constant $\mathbf{K}$. It means that the system as a whole oscillates and we find the corresponding frequencies and amplitudes.

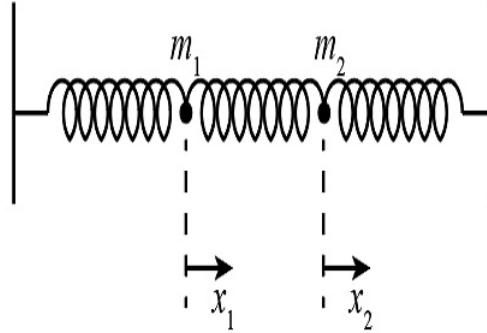


Figure 10: Two mass points connected by a spring of spring constant k_{12} (middle spring). The leftmost and rightmost strings have identical spring constant k and are attached to a rigid wall.

Normal Frequencies

In trying to solve [\(68\)](#) it is reasonable to assume that there could be solutions where both the masses move in oscillatory motions with same frequency but different amplitudes. We assume the solutions of the type

$$x_1(t) = Ae^{i\omega t}, \quad x_2(t) = Be^{i\omega t}. \quad (69)$$

The ω is the frequency to be determined and A, B are complex amplitudes that can be determined from the initial conditions. Note that the assumed solutions have complex parts which are unphysical. The actual motion must be given by real solutions only, for example $x_1(t) = A \cos(\omega t - \delta_1)$ $x_2(t) = A \cos(\omega t - \delta_2)$, where $\delta_{1,2}$ are phases. The reason for choosing exponential solutions is that it lets us combine the two solutions into a column due to same time dependance of the type $e^{i\omega t}$

$$\mathbf{X} = \begin{pmatrix} A \\ B \end{pmatrix} e^{i\omega t}. \quad (70)$$

Substituting the solutions (70) in to the EoM (68) leads to

$$-\omega^2 \mathbf{M} \mathbf{a} e^{i\omega t} + \mathbf{K} \mathbf{a} e^{i\omega t} = 0, \quad \text{or} \quad (-\omega^2 \mathbf{M} + \mathbf{K}) \mathbf{a} = 0. \quad (71)$$

where $\mathbf{a} = \begin{pmatrix} A \\ B \end{pmatrix}$. The equation (71) is known as the generalized eigenvalue equation. A non-trivial solution exist if the determinant of the matrix $-\omega^2 \mathbf{M} + \mathbf{K}$ vanishes, *i.e.*,

$$\begin{vmatrix} k + k_{12} - m\omega^2 & -k_{12} \\ -k_{12} & k + k_{12} - m\omega^2 \end{vmatrix} = 0,$$

which yields

$$(k + k_{12} - m\omega^2)^2 - k_{12}^2 = 0,$$

Solving for ω we get

$$\omega = \sqrt{\frac{k + k_{12} \pm k_{12}}{m}}.$$

So we have two *eigenfrequencies*

$$\omega_1 = \sqrt{\frac{k + 2k_{12}}{m}}, \quad \omega_2 = \sqrt{\frac{k}{m}}. \quad (72)$$

The eigenfrequencies are also known as the *normal frequencies* of the system. Looking back at the equation (68) and the solution (70), we understand that the entire system oscillates with these two frequencies. Since the system has only two degrees of freedom there are only two normal modes. If the system starts oscillating with any one of these two frequencies, it will remain in that mode for ever.

Normal Modes

Corresponding to each of the frequencies, we investigate the amplitude of solutions (70).

First Normal Mode: Let

$$\omega = \omega_1 = \sqrt{\frac{k + 2k_{12}}{m}}.$$

Substituting in equation (71) yields

$$\begin{pmatrix} k + k_{12} - k - 2k_{12} & -k_{12} \\ -k_{12} & k + k_{12} - k - 2k_{12} \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = 0,$$

which yields that $A = -B$, *i.e.*, the two masses oscillate with the same amplitude but their motions are out of phase. This is the first normal mode and the solution can be written as

$$\mathbf{X} = A_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i\omega_1 t}. \quad (73)$$

where A_1 is arbitrary constant to be fixed from initial conditions.

Second Normal Mode: Let

$$\omega = \omega_2 = \sqrt{\frac{k}{m}}.$$

Substituting in equation (71) yields

$$\begin{pmatrix} k + k_{12} - k & -k_{12} \\ -k_{12} & k + k_{12} - k \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = 0,$$

which yields that $A = B$, i.e., the two masses oscillate with the same amplitude and their motions are in phase. This is the second normal mode

$$\mathbf{X} = B_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{i\omega_2 t}. \quad (74)$$

where B_1 can be fixed from initial conditions.

Normal Coordinates

Since the normal modes are independent of each other, one can write the general solution of the problem as

$$\mathbf{X} = A_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i\omega_1 t} + A_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{i\omega_2 t}. \quad (75)$$

To be more general we can write

$$\mathbf{X} = A_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i(\omega_1 t - \delta_1)} + A_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{i(\omega_2 t - \delta_2)}. \quad (76)$$

The actual motion is given by the real part of the solution

$$\mathbf{X} = A_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cos(\omega_1 t - \delta_1) + A_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cos(\omega_2 t - \delta_2). \quad (77)$$

The motion of the coordinates $x_1(t)$ and $x_2(t)$ are not easy to visualize. However, one can introduce normal coordinates as a superposition of $x_{1,2}$ whose motions are easy to describe

$$\eta_1 = \frac{1}{2}(x_1 - x_2), \quad \eta_2 = \frac{1}{2}(x_1 + x_2).$$

Taking a similar linear superposition of the two equations EoMs (67) we get

$$m\ddot{\eta}_1 + (k + 2k_{12})\eta_1 = 0, \quad m\ddot{\eta}_2 + k\eta_2 = 0, \quad (78)$$

These equations show that the solutions of η_1 oscillates frequency $\omega_1 = \sqrt{(k + 2k_{12})/m}$ and η_2 oscillates with frequency $\omega_2 = \sqrt{k/m}$. It is easy to see that for the first normal mode when $\omega = \omega_1$

$$\eta_1 = A_1 e^{i\omega_1 t}, \quad \eta_2 = 0,$$

and for the second normal mode

$$\eta_1 = 0, \quad \eta_2 = B_1 e^{-i\omega_2 t}.$$

The η_1 and η_2 are known as the normal modes and the most general solutions can be written as

$$\eta_1(t) = E_1 e^{i\omega_1 t} + E_2 e^{i\omega_1 t}, \quad \eta_2(t) = F_1 e^{i\omega_2 t} + F_2 e^{i\omega_2 t} \quad (79)$$

The four constants $E_{1,2}$ and $F_{1,2}$ can be determined from the initial conditions of the system. Let us chose as initial conditions $x_1(0) = -x_2(0)$ and $\dot{x}_1 = -\dot{x}_2$. In this case we see that $\eta_2(0) = 0$ and $\dot{\eta}_2(0) = 0$ which gives $F_1 = F_2 = 0$, i.e., $\eta_2(t) = 0$ for all times. This is the antisymmetrical mode where the two particles oscillate in the opposite directions (*out of phase*) with frequency ω_1 . On the other hans for the initial conditions $x_1(0) = x_2(0)$ and $\dot{x}_1 = \dot{x}_2$ we find that $\eta_1(t) = 0$ for all the times t – two masses oscillate *in phase* with frequency ω_2 . This is the symmetrical solution. The fact that the antisymmetrical mode has a higher frequency is a general result.

5.2 Three coupled pendulums

Please refer to Taylor's book. There are three identical pendulums of mass m . The Lagrangian is

$$L = \frac{1}{2}ml^2 (\dot{\theta}_1^2 + \dot{\theta}_2^2 + \dot{\theta}_3^2) - \frac{1}{2}mgl (\theta_1^2 + \theta_2^2 + \theta_3^2) - \frac{1}{2}kl^2 [(\theta_2 - \theta_1)^2 + (\theta_3 - \theta_2)^2]. \quad (80)$$

Defining $\omega_0^2 = g/\ell$ and $\alpha = k/m$ the equations of motions are

$$\ddot{\theta}_1 + \omega_0^2 \theta_1 + \alpha(\theta_1 - \theta_2) = 0, \quad (81)$$

$$\ddot{\theta}_2 + \omega_0^2 \theta_2 + \alpha(2\theta_2 - \theta_1 - \theta_3) = 0, \quad (82)$$

$$\ddot{\theta}_3 + \omega_0^2 \theta_3 + \alpha(\theta_3 - \theta_2) = 0. \quad (83)$$

which can be written in matrix form as

$$\ddot{\boldsymbol{\theta}} + \begin{pmatrix} \omega_0^2 + \alpha & -\alpha & 0 \\ -\alpha & \omega_0^2 + 2\alpha & -\alpha \\ 0 & -\alpha & \omega_0^2 + \alpha \end{pmatrix} \boldsymbol{\theta} = 0, \quad (84)$$

where

$$\boldsymbol{\theta} = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{pmatrix}. \quad (85)$$

Note that the equation (5.2) is similar to (68). Without repeating the calculations of finding the normal frequencies and the normal modes, I simply write the results here – please see the book of Taylor for detailed derivations. The normal modes are

$$\omega_1^2 = \omega_0^2, \omega_2^2 = \omega_0^2 + k, \omega_3^2 = \omega_0^2 + 3k \quad (86)$$

and the normalized eigen vectors are $(1, 1, 1)/\sqrt{3}$, $(1, 0, -1)/\sqrt{2}$, $(1, -2, 1)/\sqrt{6}$. So the general solution is

$$\begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \text{Re}(C_1 e^{i\omega_1 t}) + \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \text{Re}(C_2 e^{i\omega_2 t}) + \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \text{Re}(C_3 e^{i\omega_3 t}) \quad (87)$$

where C_i are arbitrary constants. The three column vectors $\mathbf{a}_1 = (1, 1, 1)/\sqrt{3}$, $\mathbf{a}_2 = (1, 0, -1)/\sqrt{2}$, $\mathbf{a}_3 = (1, -2, 1)/\sqrt{6}$ are linearly independent vectors.

As explained in the Taylor's book the method of finding normal coordinates is as follows: Any arbitrary solution can be written as a linear combinations of the vectors $\mathbf{a}_i$ as

$$\boldsymbol{\vartheta} = \sum_i \xi_i \mathbf{a}_i, \quad (88)$$

and it defines the normal coordinates ξ_i . It means that $\boldsymbol{\vartheta}$ is a solution of the equation. If you substitute the $\boldsymbol{\vartheta}$ in we will find that the equations of motions of ξ_i by single frequency as

$$\ddot{\xi}_i = -\omega_i^2 \xi_i. \quad (89)$$

In equation (87) one can identify that $\xi_i = \text{Re}(C_1 e^{i\omega_1 t})$. Therefore

$$\begin{aligned} \theta_1 &= \xi_1/\sqrt{3} + \xi_2/\sqrt{2} + \xi_3/\sqrt{6} \\ \theta_2 &= \xi_1/\sqrt{3} - 2\xi_3/\sqrt{6} \\ \theta_3 &= \xi_1/\sqrt{3} - \xi_2/\sqrt{2} + \xi_3/\sqrt{6} \end{aligned} \quad (90)$$

Substituting in the Lagrangian we get

$$L = \frac{1}{2} [\dot{\xi}_1^2 - \omega_0^2 \xi_1^2] + \frac{1}{2} [\dot{\xi}_2^2 - (\omega_0^2 + k) \xi_2^2] + \frac{1}{2} [\dot{\xi}_3^2 - (\omega_0^2 + 3k) \xi_3^2] \quad (91)$$

This is three independent Lagrangian for the three normal coordinates $\xi_{1,2,3}$. The set of equations (90) can be solved for ξ_i to obtain the expressions of normal coordinates in terms of the generalized coordinates.